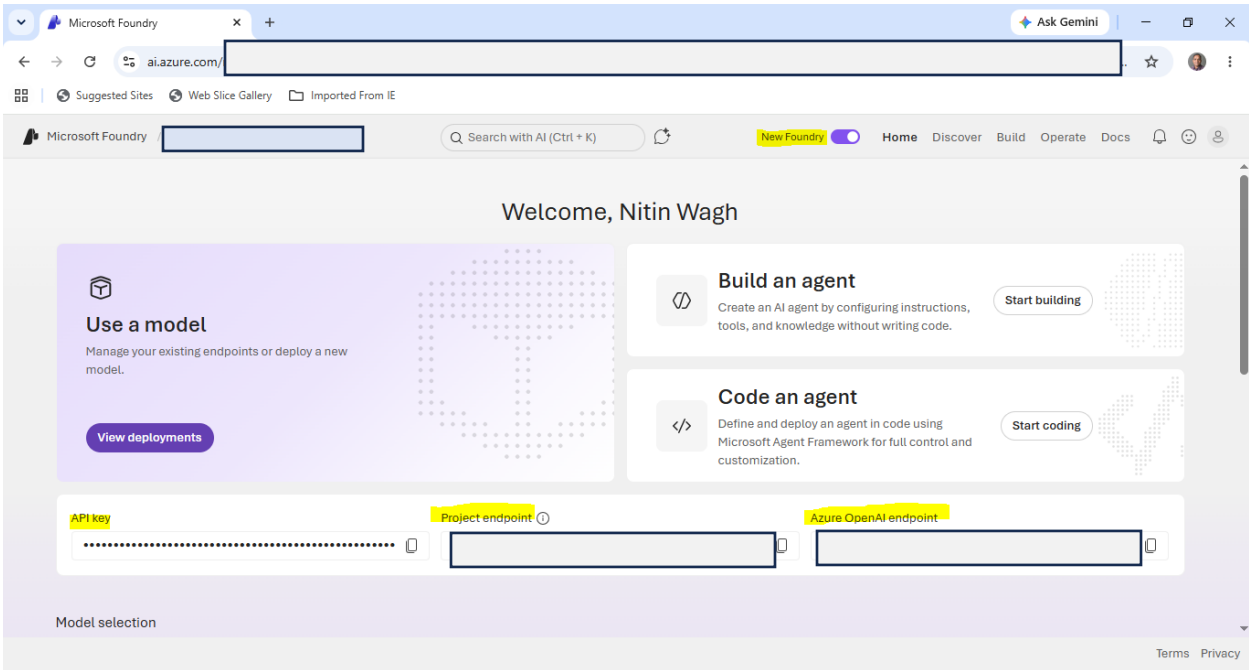


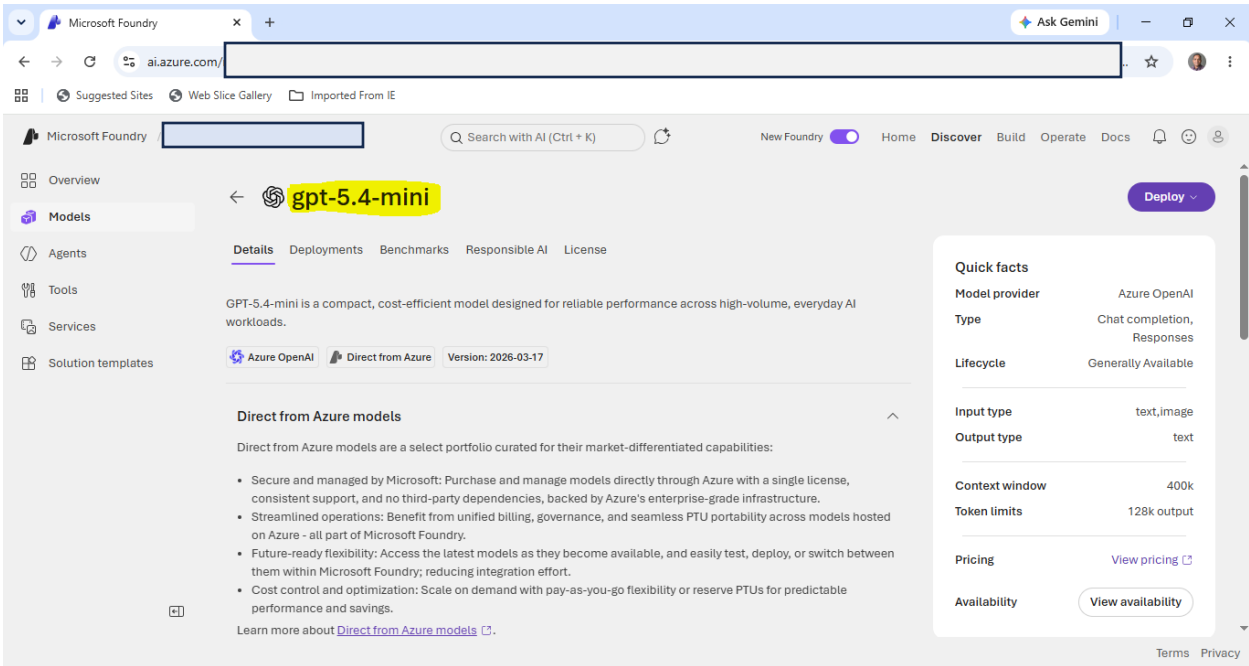
Azure Text Analysis Project

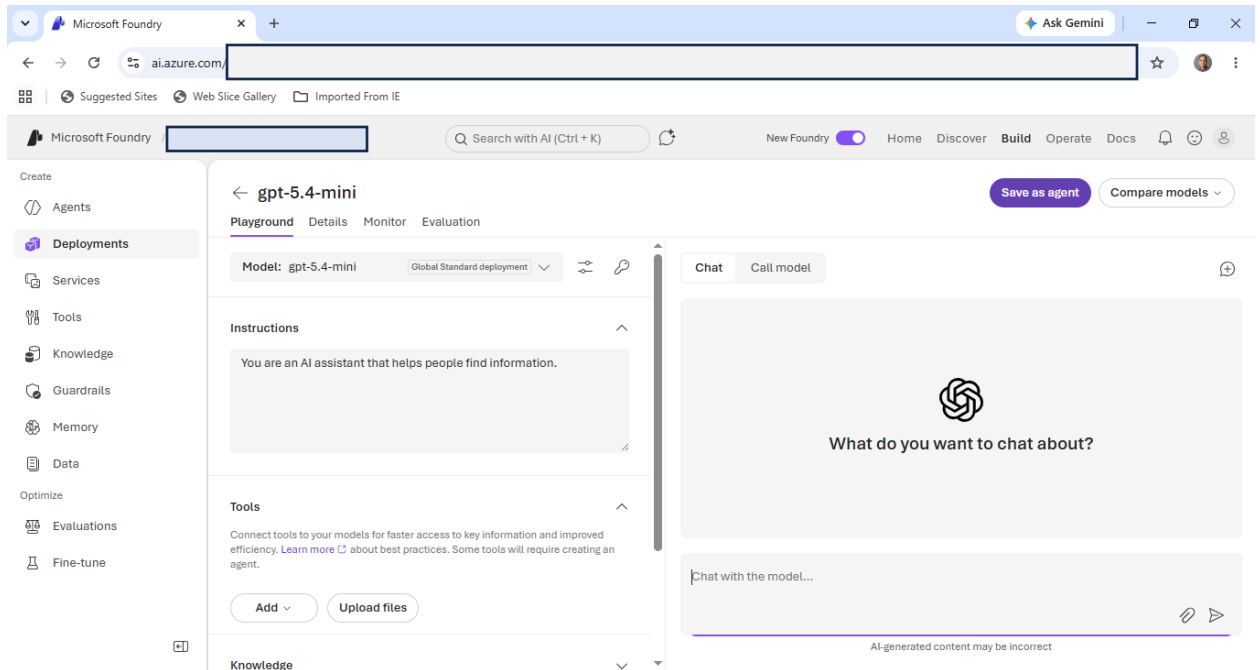
Initial Setup:

1. Project Creation



2. Model Selection & Deployment





Code Output:

1. Sentiment Analysis

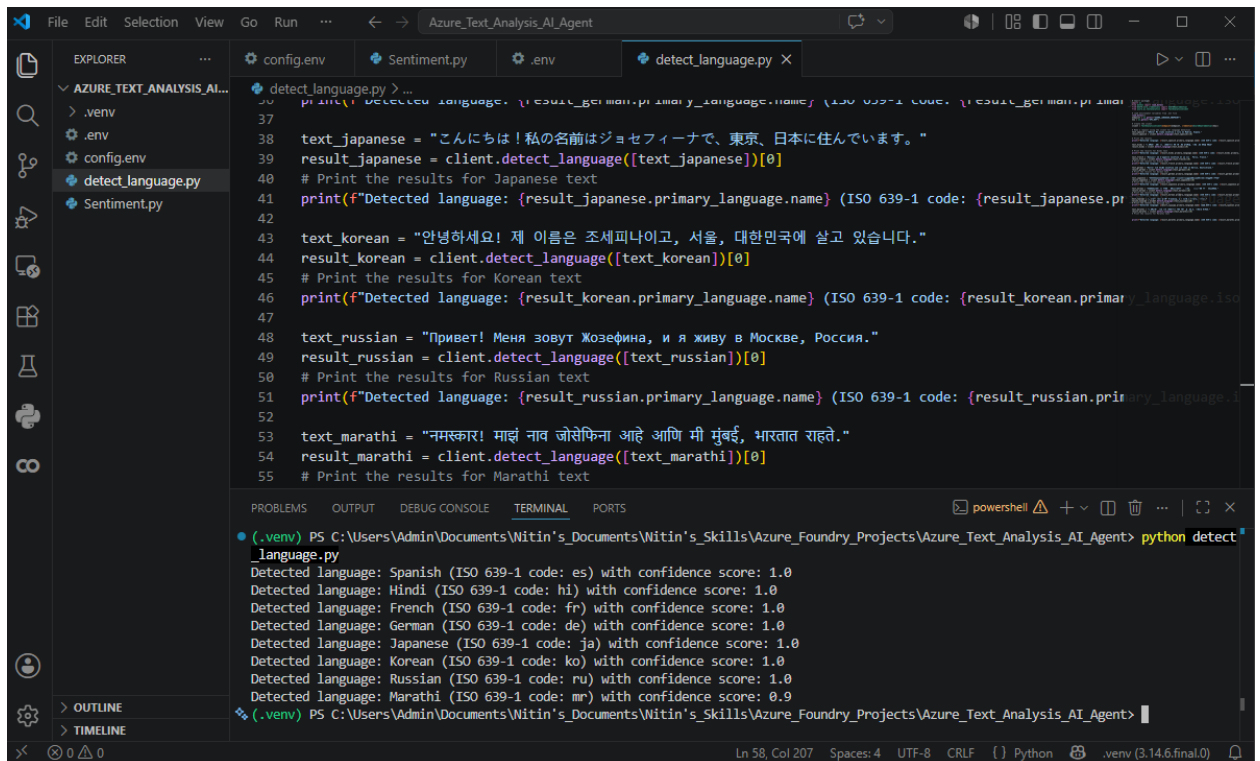
```

File Edit Selection View Go Run ... Azure_Text_Analysis_AI_Agent
EXPLORER
AZURE_TEXT_ANALYSIS_AI...
  .venv
  .env
  config.env
  detect_language.py
  Sentiment.py
Sentiment.py
0 load_dotenv()
7 endpoint = os.getenv("AZURE_OPENAI_ENDPOINT")
8 api_key = os.getenv("API_KEY")
9 deployment_name = os.getenv("MODEL_DEPLOYMENT_NAME")
10
11 # Create the client object
12 client = OpenAI(
13     base_url=endpoint,
14     api_key=api_key
15 )
16
17 # Make a request using the client
18 message = client.responses.create(
19     model=deployment_name,
20     input="What is the sentiment of the following text: 'I love programming, but sometimes it can be frustrating.'"
21 )
22
23 # Print the results
24 print(f"Sentiment: {message.output[0]}")

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
(.venv) PS C:\Users\Admin\Documents\Witin's_Documents\Witin's_Skills\Azure_Foundry_Projects\Azure_Text_Analysis_AI_A
gent> python sentiment.py
Sentiment: ResponseOutputMessage(id='msg_00d967209c5e4788006a57bcb50e5081939fc61a771770faf4', content=[ResponseOutputText(annotations=[
], text='The sentiment is **mixed/neutral leaning positive**.\n\n**Positive:** "I love programming"\n\n**Negative:** "sometimes it ca
n be frustrating"\n\nOverall, it expresses both enthusiasm and frustration, so it's **mixed sentiment**.', type='output_text', logprobs
=[])], role='assistant', status='completed', type='message', phase='final_answer')
(.venv) PS C:\Users\Admin\Documents\Witin's_Documents\Witin's_Skills\Azure_Foundry_Projects\Azure_Text_Analysis_AI_Agent>

```

2. Language Detection



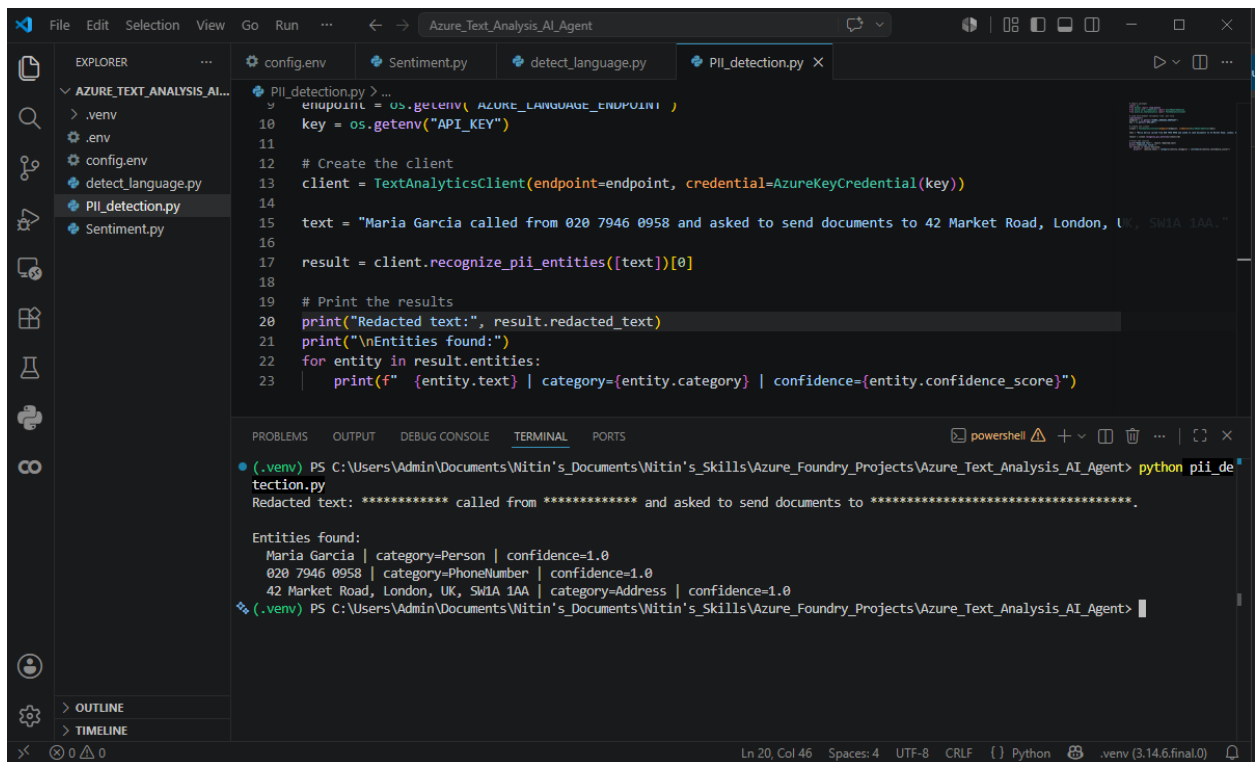
The screenshot shows the Visual Studio Code editor with the file `detect_language.py` open. The script uses the `azure-text-analytics` library to detect the language of several text samples. The terminal output shows the results for each sample, including the detected language, ISO 639-1 code, and confidence score.

```
36 print(f"Detected language: {result_japanese.primary_language.name} (ISO 639-1 code: {result_japanese.primary_language.iso_639_1_code}) with confidence score: {result_japanese.confidence_score}")
37
38 text_japanese = "こんにちは！私の名前はジョセフィーナで、東京、日本に住んでいます。"
39 result_japanese = client.detect_language([text_japanese])[0]
40 # Print the results for Japanese text
41 print(f"Detected language: {result_japanese.primary_language.name} (ISO 639-1 code: {result_japanese.primary_language.iso_639_1_code}) with confidence score: {result_japanese.confidence_score}")
42
43 text_korean = "안녕하세요! 제 이름은 조세피나이고, 서울, 대한민국에 살고 있습니다."
44 result_korean = client.detect_language([text_korean])[0]
45 # Print the results for Korean text
46 print(f"Detected language: {result_korean.primary_language.name} (ISO 639-1 code: {result_korean.primary_language.iso_639_1_code}) with confidence score: {result_korean.confidence_score}")
47
48 text_russian = "Привет! Меня зовут Жозефина, и я живу в Москве, Россия."
49 result_russian = client.detect_language([text_russian])[0]
50 # Print the results for Russian text
51 print(f"Detected language: {result_russian.primary_language.name} (ISO 639-1 code: {result_russian.primary_language.iso_639_1_code}) with confidence score: {result_russian.confidence_score}")
52
53 text_marathi = "नमस्कार! माझा नाव जोसेफिना आहे आणि मी मुंबई, भारतात राहतो."
54 result_marathi = client.detect_language([text_marathi])[0]
55 # Print the results for Marathi text
```

Terminal Output:

```
(.venv) PS C:\Users\Admin\Documents\Witin's_Documents\Witin's_Skills\Azure_Foundry_Projects\Azure_Text_Analysis_AI_Agent> python detect_language.py
Detected language: Spanish (ISO 639-1 code: es) with confidence score: 1.0
Detected language: Hindi (ISO 639-1 code: hi) with confidence score: 1.0
Detected language: French (ISO 639-1 code: fr) with confidence score: 1.0
Detected language: German (ISO 639-1 code: de) with confidence score: 1.0
Detected language: Japanese (ISO 639-1 code: ja) with confidence score: 1.0
Detected language: Korean (ISO 639-1 code: ko) with confidence score: 1.0
Detected language: Russian (ISO 639-1 code: ru) with confidence score: 1.0
Detected language: Marathi (ISO 639-1 code: mr) with confidence score: 0.9
(.venv) PS C:\Users\Admin\Documents\Witin's_Documents\Witin's_Skills\Azure_Foundry_Projects\Azure_Text_Analysis_AI_Agent>
```

3. PII Detection



The screenshot shows the Visual Studio Code editor with the file `pii_detection.py` open. The script uses the `azure-text-analytics` library to detect PII entities in a text sample. The terminal output shows the redacted text and the detected entities with their categories and confidence scores.

```
9 endpoint = os.getenv('AZURE_LANGUAGE_ENDPOINT')
10 key = os.getenv('API_KEY')
11
12 # Create the client
13 client = TextAnalyticsClient(endpoint=endpoint, credential=AzureKeyCredential(key))
14
15 text = "Maria Garcia called from 020 7946 0958 and asked to send documents to 42 Market Road, London, UK, SW1A 1AA."
16
17 result = client.recognize_pii_entities([text])[0]
18
19 # Print the results
20 print("Redacted text:", result.redacted_text)
21 print("\nEntities found:")
22 for entity in result.entities:
23     print(f" {entity.text} | category={entity.category} | confidence={entity.confidence_score}")
```

Terminal Output:

```
(.venv) PS C:\Users\Admin\Documents\Witin's_Documents\Witin's_Skills\Azure_Foundry_Projects\Azure_Text_Analysis_AI_Agent> python pii_detection.py
Redacted text: ***** called from ***** and asked to send documents to *****

Entities found:
Maria Garcia | category=Person | confidence=1.0
020 7946 0958 | category=PhoneNumber | confidence=1.0
42 Market Road, London, UK, SW1A 1AA | category=Address | confidence=1.0
(.venv) PS C:\Users\Admin\Documents\Witin's_Documents\Witin's_Skills\Azure_Foundry_Projects\Azure_Text_Analysis_AI_Agent>
```